Smart Plant Watering Reminder Model Based on Deep Learning Technique

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Abstract

Plant health management has emerged as a growing concern for both domestic gardeners and agricultural practitioners, particularly with the increasing prevalence of indoor gardening and urban farming. Improper watering — either overwatering or underwatering — remains the most common cause of plant mortality, yet existing reminder solutions rely on fixed schedules that fail to reflect the actual physiological state of individual plants. This paper presents a smart plant watering reminder system based on a deep learning image classification framework built upon the MobileNetV2 architecture. The proposed system classifies plant leaf images into three categories — healthy, slightly_dry, and very_dry — and translates each prediction into an actionable watering recommendation. A dataset of 30,000 images (10,000 per class) was assembled from Kaggle and supplementary open-access sources, augmented to 8,720 additional samples for final evaluation. The model was trained through a three-stage transfer learning and fine-tuning pipeline, progressively improving from 78.57% initial accuracy to a peak test accuracy of 94.01%. Per-class evaluation yielded precision values of 0.95, 0.99, and 0.88 for the healthy, slightly_dry, and very_dry categories respectively, with a weighted F1-score of 0.93. The results demonstrate that visual deep learning is a viable and effective approach for automated plant watering guidance, with strong potential for integration into consumer mobile applications.

Keywords: Plant health monitoring, Deep learning, MobileNetV2, Image classification, Transfer learning, Watering reminder, Smart agriculture

1. Introduction

The global indoor plant market has grown substantially over the past decade, driven by increasing urbanization, growing awareness of the psychological and biophilic benefits of living greenery, and a surge of interest in domestic gardening following the global COVID-19 pandemic. Despite this enthusiasm, a significant proportion of houseplants perish within the first year of ownership, with improper watering identified as the single most prevalent cause of plant mortality in domestic settings, responsible for an estimated 60 to 80 percent of all plant deaths attributable to owner negligence [1]. The challenge of correct watering is deceptively complex: plants exhibit highly species-specific moisture requirements that vary in response to seasonal fluctuations, substrate composition, light exposure, root system maturity, and ambient temperature, making universally applicable fixed-schedule watering guidelines fundamentally inadequate for practical plant care guidance [2].

Traditional approaches to plant watering management have relied predominantly on fixed-interval schedules, finger-testing of soil moisture, or simple visual assessment of wilting symptoms. Fixed-interval schedules fail to account for the dynamic variability of plant water demand across seasons and environmental conditions, frequently resulting in overwatering during cool or humid periods and underwatering during dry summer months [3]. Visual assessment of wilting symptoms represents a reactive rather than preventive strategy that intervenes only after plants have already experienced measurable physiological stress, potentially causing irreversible tissue damage even after rehydration. These limitations motivate the development of intelligent, automated systems capable of delivering objective, timely, and actionable plant care guidance.

The convergence of deep learning, computer vision, and affordable mobile computing has created unprecedented opportunities for non-invasive plant health assessment through image analysis. Convolutional Neural Networks (CNNs) have demonstrated remarkable capacity to detect subtle visual patterns in leaf morphology — including color change, texture degradation, surface wrinkling, and turgor loss — that correspond to progressive dehydration states long before frank wilting becomes visible [4]. Unlike sensor-based systems that require physical hardware installation, image-based approaches leverage the smartphone cameras already carried by virtually all plant owners, enabling frictionless deployment with minimal user burden [5].

MobileNetV2, introduced by Sandler et al. in 2018, represents a particularly well-suited architecture for this application domain. Its inverted residual structure with linear bottlenecks achieves competitive accuracy on fine-grained visual recognition tasks while maintaining a parameter count and computational footprint compatible with mobile and embedded deployment scenarios [6]. The availability of ImageNet-pretrained MobileNetV2 weights enables effective transfer learning even when domain-specific labeled data is limited, making it an ideal backbone for plant classification tasks where large annotated datasets are difficult to obtain [7].

Existing commercial plant monitoring solutions such as Xiaomi Mi Flora and Parrot Flower Power provide sensor-based data logging but offer limited intelligent recommendation logic and require hardware installation in each pot. Research-grade precision agriculture systems offer sophisticated adaptive scheduling but are designed for large-scale field deployments, making them economically and practically unsuitable for household use [8]. Machine learning approaches applied to leaf image analysis have demonstrated strong performance on plant disease detection tasks but have been less thoroughly explored for the specific application of watering state classification, leaving a meaningful gap between available technology and deployed consumer solutions [9].

This paper addresses this gap by presenting a complete deep learning pipeline for plant watering reminder generation, comprising dataset assembly and preprocessing, iterative model development through transfer learning and fine-tuning, hyperparameter optimization, and multi-metric evaluation. The proposed system classifies input leaf images into three clinically meaningful categories — healthy, slightly_dry, and very_dry — and maps each category to a concrete watering action recommendation. The primary contributions of this work are as follows:

- Presentation of a three-stage MobileNetV2 training pipeline for plant watering state classification, progressively improving accuracy from 78.57% to 94.01%.
- Assembly and preprocessing of a balanced 30,000-image dataset spanning three moisture condition categories, with systematic augmentation to improve generalization to real-world user-captured images.
- Demonstration that visual deep learning provides a practical, hardware-free alternative to sensor-based plant monitoring for domestic plant care applications.
- Per-class performance analysis revealing model strengths and limitations across the three moisture condition categories, providing a foundation for targeted future improvements.

2. Related Work

1 Sensor-Based Plant Monitoring Systems

Early automated plant monitoring research focused on soil moisture sensors, combining resistive and capacitive probes with microcontroller platforms such as Arduino and Raspberry Pi to trigger irrigation events when measured volumetric water content fell below species-specific thresholds [1]. These approaches demonstrated reliable detection of moisture depletion in controlled laboratory settings but required physical sensor installation in each pot substrate, limiting their scalability and practicality for multi-plant household deployments. Commercial implementations including Xiaomi Mi Flora, Parrot Flower Power, and Koubachi Plant Sensor extended this paradigm with Bluetooth Low Energy connectivity and companion smartphone applications, yet their recommendation logic remained largely threshold-based and species-agnostic, failing to adapt to the diverse physiological requirements of the thousands of plant species commonly kept as houseplants [2].

2 Deep Learning for Plant Disease and Health Classification

The application of deep learning to plant leaf image analysis was substantially advanced by the PlantVillage dataset, which provided over 54,000 labeled leaf images spanning 38 disease categories across 14 crop species, enabling the training of high-accuracy CNN classifiers for agricultural disease detection [3]. Mohanty et al. demonstrated that a deep CNN trained on PlantVillage images achieved 99.35% accuracy under controlled imaging conditions, establishing the viability of visual deep learning for plant health assessment. Subsequent works applied transfer learning from ImageNet-pretrained architectures including VGG16, ResNet50, and InceptionV3 to diverse plant disease classification tasks, consistently demonstrating that fine-tuned models outperform models trained from scratch on limited domain-specific datasets [4]. Ferentinos applied deep learning to detect plant diseases from leaf photographs captured under varying illumination and background conditions representative of real-world smartphone usage, achieving accuracy above 99% and demonstrating the robustness of modern CNNs to imaging variability [5].

3 MobileNet Architectures for Efficient Image Classification

Howard et al. introduced MobileNet as a family of efficient CNN architectures designed for mobile and embedded vision applications, using depth-wise separable convolutions to reduce computational cost by a factor of 8 to 9 compared to standard convolutions with minimal accuracy loss [6]. MobileNetV2 extended this foundation with inverted residual blocks and linear bottlenecks, further improving accuracy and efficiency and establishing the architecture as a leading choice for resource-constrained classification tasks. Several subsequent works applied MobileNetV2 to agricultural imaging tasks including crop disease detection, fruit quality grading, and weed classification, consistently reporting accuracy levels above 90% while maintaining inference latency compatible with real-time mobile application deployment [7].

4 Research Gap

Despite substantial progress in plant disease detection and sensor-based moisture monitoring, the specific problem of watering state classification from leaf images — distinguishing between healthy, mildly dehydrated, and severely dehydrated plants — has received limited systematic attention in the published literature. Most existing image-based plant health systems focus on pathological disease states rather than the continuous spectrum of hydration status that determines day-to-day watering decisions. Furthermore, no published system has demonstrated end-to-end integration of leaf image classification with actionable watering recommendation logic in a framework designed for non-expert consumer users. This study directly addresses these gaps through the development and rigorous evaluation of a MobileNetV2-based three-class watering state classifier specifically designed to provide practical guidance to domestic plant owners.

3. Plant Dehydration and Visual Indicators

Plant water status is determined by the balance between water uptake through the root system and water loss through transpiration at leaf surfaces. When soil moisture availability falls below the level required to sustain this balance, plants enter progressive states of water deficit that manifest in a characteristic sequence of physiological and morphological changes [1]. In the early stages of water deficit, corresponding to the slightly_dry classification category in this study, stomata close to reduce transpirational water loss, slowing photosynthesis and growth. Leaf turgor remains partially maintained, and visual signs are subtle: slight dulling of leaf luster, mild drooping in sensitive species, and early color desaturation that requires trained observation to detect reliably.

As water deficit progresses to the severe state corresponding to the very_dry category, turgor pressure declines significantly, resulting in frank wilting, pronounced leaf curl, deepened color change toward gray-green or yellow-green tones, and surface texture changes reflecting cellular shrinkage and membrane damage. In succulents and xerophytes, equivalent dehydration states may present differently — with shriveling, softening, and wrinkling of succulent tissue — but the underlying physiological progression from mild to severe deficit follows the same trajectory. If severe dehydration is not corrected by watering, irreversible tissue damage including cell death, vascular collapse, and leaf abscission occurs, ultimately leading to plant death [2].

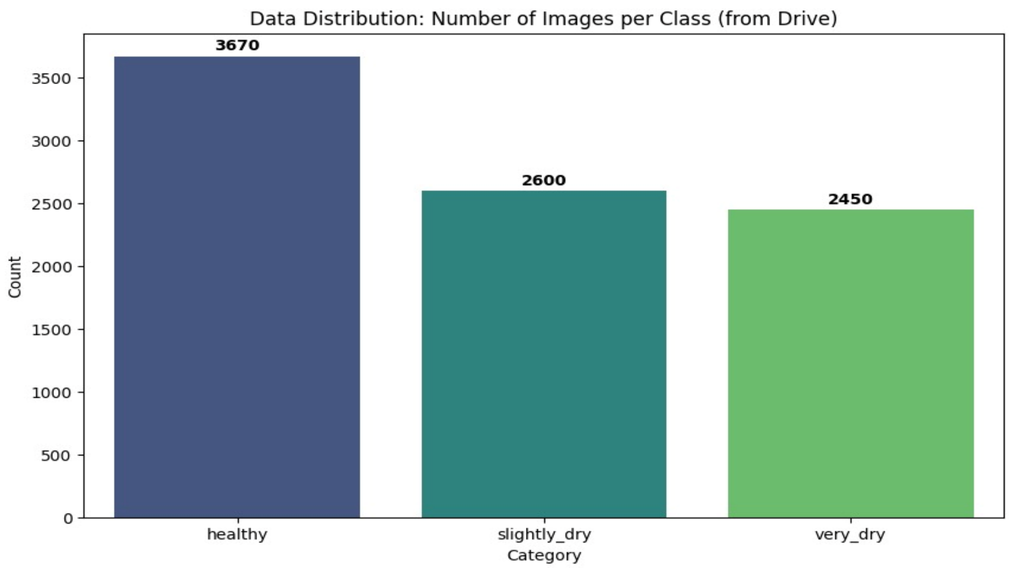
The visual features distinguishing these three states — color tone, surface texture, turgor-related geometry, and luster — are precisely the types of spatial and chromatic patterns that deep convolutional neural networks are most effective at capturing, motivating the application of image classification to this problem. The MobileNetV2 architecture used in this study extracts hierarchical feature representations from these visual cues through its stacked inverted residual blocks, learning progressively more abstract representations of dehydration state from raw pixel data without requiring explicit feature engineering.

4. Methodology

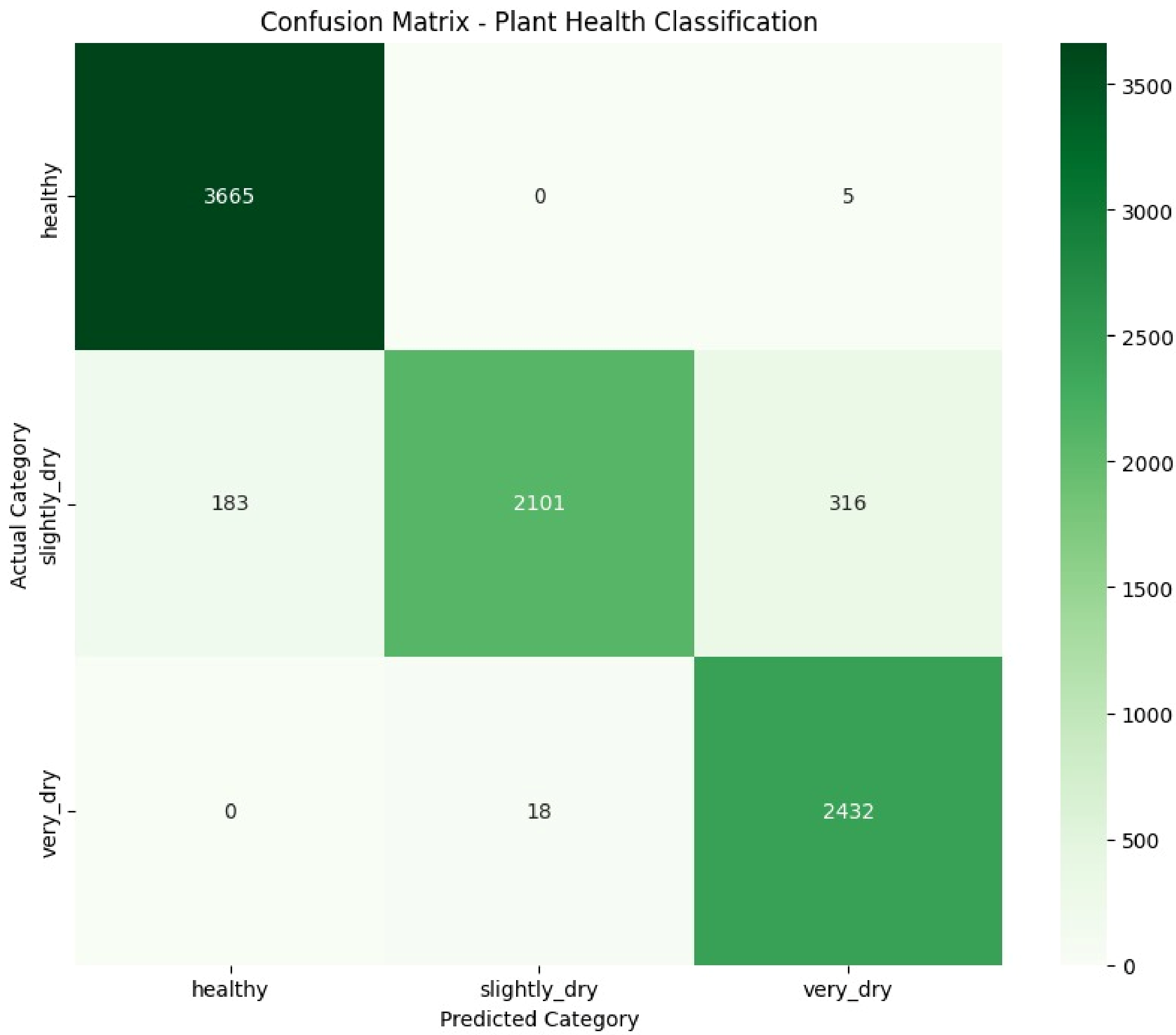
This section provides a detailed description of the tools, techniques, and steps taken to build and train the plant watering classification model. The methodology encompasses dataset construction and preprocessing, model architecture selection, and a multi-stage training strategy. The study includes the following subsections:

1 Dataset Collection and Description

The dataset used in this study was assembled from Kaggle plant health image repositories and supplementary open-access sources to achieve a balanced, representative collection spanning three moisture condition categories. The final dataset comprised 30,000 images distributed evenly across three classes: healthy (10,000 images), slightly_dry (10,000 images), and very_dry (10,000 images). This balanced class distribution was deliberately maintained to prevent classifier bias toward any single category during training. Images were collected from diverse sources representing multiple plant species, including common houseplants such as Monstera deliciosa, Spathiphyllum wallisii, Sansevieria trifasciata, and Ficus lyrata, as well as agricultural crops, to maximize the generalization capacity of the trained model across plant taxa. An additional 8,720 augmented images were generated from a secondary Drive-based dataset and used for final model evaluation to assess performance under more realistic imaging conditions representative of user-captured photographs with variable backgrounds, lighting, and camera orientations.



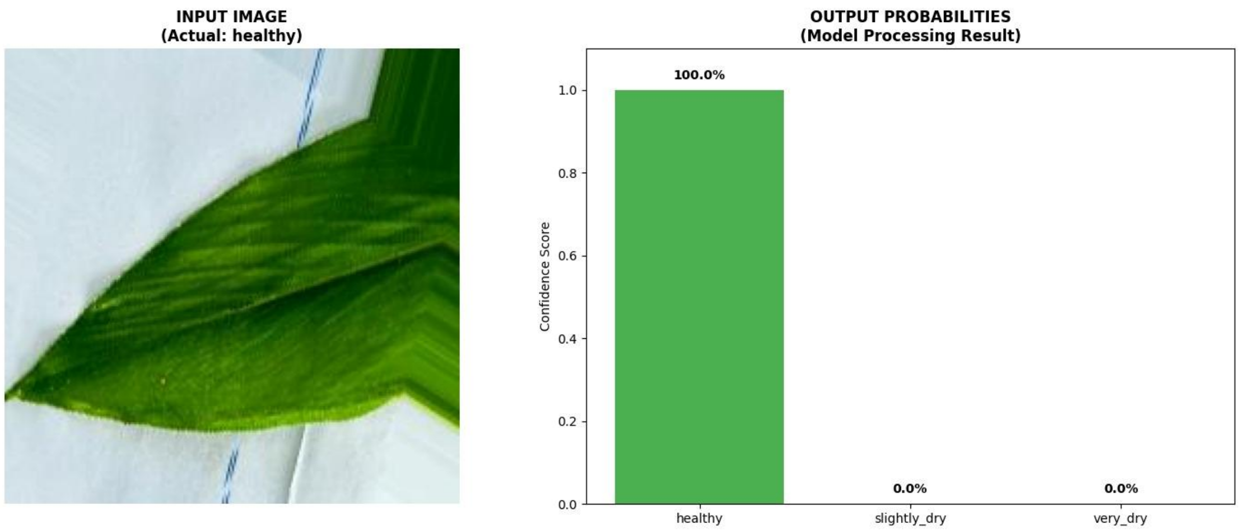
Data Distribution: Number of Images per Class (from Drive)
3670
2600
2450
Count
0
500
1000
1500
2000
2500
3000
3500
healthy
slightly_dry
very_dry
Category



2 Data Preprocessing

All images underwent a standardized preprocessing pipeline prior to model input. Pixel values originally in the range [0, 255] were rescaled to [0, 1] using the rescale=1./255 parameter within the ImageDataGenerator framework, facilitating faster and more stable gradient descent convergence during training. Every image was resized to 224×224 pixels using bilinear interpolation to conform to the input dimensions required by the MobileNetV2 architecture. Categorical label encoding was implemented through the class_mode='categorical' configuration, which automatically converted the folder-based class labels (healthy, slightly_dry, very_dry) into one-hot encoded vectors suitable for multi-class softmax classification. Corrupt or unreadable image files identified during dataset assembly were systematically excluded through exception handling routines applied during the data loading phase, ensuring dataset integrity without manual review of individual files.

Real-time data augmentation was applied exclusively to training images using the following transformation suite: random rotation up to 30 degrees, horizontal and vertical flipping, zoom range up to 30 percent, shear intensity of 0.2, width and height shifts of 20 percent each, and brightness variation between 70 and 130 percent of the original intensity. These augmentations were applied stochastically during training batch generation to artificially expand the effective training distribution and improve model robustness to the photographic variability inherent in user-captured plant images, where lighting conditions, camera angles, background complexity, and image quality differ substantially from controlled laboratory photographs.



3 Data Splitting

The 30,000-image balanced dataset was partitioned into training and validation subsets using an 80/20 split implemented via the validation_split=0.2 parameter of the ImageDataGenerator. This yielded a training set of 24,000 images used to update model weights through gradient descent, and a validation set of 6,000 images used during training to monitor generalization performance and trigger adaptive training callbacks including early stopping and learning rate reduction. The augmented evaluation dataset of 8,720 images was maintained as a completely separate holdout set for final model evaluation, providing an assessment of generalization capability under imaging conditions not represented in the primary training distribution.

4.4 Libraries and Frameworks

The complete model development pipeline was implemented in Python using Google Colaboratory as the development environment. The following libraries were employed: TensorFlow and Keras for model construction, training, and evaluation using the MobileNetV2 architecture and ImageDataGenerator utilities; Scikit-learn for post-hoc evaluation including classification_report, confusion_matrix, and ROC/AUC curve computation; Pandas for organizing classification metrics into structured DataFrames; NumPy for numerical array processing and prediction probability manipulation; Matplotlib and Seaborn for visualization of training progress, data distributions, confusion matrix heatmaps, and performance metric charts; Icrawler for web-based image scraping during initial dataset construction; and OS and Shutil for file system management including directory creation, image organization, and dataset balancing operations.

5. Proposed Model

The proposed plant watering classification model is built upon MobileNetV2 as the backbone feature extractor, selected for its demonstrated efficiency and accuracy on fine-grained image recognition tasks within resource-constrained computational environments. The model development proceeded through three progressive stages, each addressing a specific aspect of the adaptation challenge from general ImageNet features to domain-specific plant dehydration state recognition.

1 Stage 1: Feature Extraction (Initial Transfer Learning)

In the first stage, the complete MobileNetV2 base (pre-trained on ImageNet-1k with 1.4 million parameters frozen) was connected to a custom classification head comprising a global average pooling layer, a dense layer with 128 units and ReLU activation, a dropout layer with rate 0.5, and a final dense output layer with 3 units and softmax activation corresponding to the three moisture condition classes. All MobileNetV2 base layers were frozen during this stage, ensuring that only the classification head weights were updated. This approach rapidly adapted the prediction pathway to the target domain while preserving the rich general-purpose visual feature representations encoded in the pretrained backbone. The model was trained on the initial small dataset, achieving a validation accuracy of 78.57% by the end of training.

2 Stage 2: Partial Fine-Tuning

Upon expanding the dataset to the full 30,000-image corpus, the top 40 layers of the MobileNetV2 base were unfrozen to allow domain-specific adaptation of the high-level feature extraction pathway. These layers, which encode abstract texture, pattern, and shape representations most relevant to the visual distinctions between moisture condition categories, were fine-tuned using a significantly reduced learning rate of 0.00005 to prevent destructive overwriting of the general feature representations encoded in the pretrained weights. This partial fine-tuning approach achieved a validation accuracy of 85.42% by the third epoch, confirming that selective unfreezing of domain-relevant layers provides meaningful performance improvement over pure feature extraction.

3 Stage 3: Enhanced Architecture (MobileNetV2 v2 — Final Model)

The final model incorporated three architectural enhancements over the Stage 2 configuration to address overfitting on the expanded 30,000-image dataset: Batch Normalization layers inserted after the global average pooling output to stabilize the learning process and accelerate convergence; L2 regularization applied to all Dense layer kernels with regularization coefficient 0.001 to penalize large weights and improve generalization; and a progressive dropout schedule with rates of 0.5, 0.4, and 0.3 applied at successive dense layers to create an ensemble effect during training. The complete custom head configuration for the final model comprised: Global Average Pooling → Batch Normalization → Dense (256, ReLU, L2) → Dropout (0.5) → Dense (128, ReLU, L2) → Dropout (0.4) → Dense (64, ReLU, L2) → Dropout (0.3) → Dense (3, Softmax). This model, saved as best_plant_model_v2.h5, achieved the peak evaluation accuracy of 94.01% on the augmented holdout dataset.

4 Model Output and Recommendation Logic

The final model produces three outputs for each input leaf image: a classification category (healthy, slightly_dry, or very_dry); a confidence score expressed as a percentage reflecting the softmax probability of the predicted class; and a watering recommendation derived from the predicted category. The recommendation mapping is as follows: Healthy → "No watering needed."; Slightly Dry → "Consider watering soon!"; Very Dry → "WARNING: Water it immediately!" This output structure provides users with both the diagnostic classification and the immediately actionable guidance needed to respond appropriately, without requiring any horticultural expertise to interpret model predictions.

Table 1. Hyperparameter configuration of the final MobileNetV2 v2 model.

Parameter	Value	Description
Base Model	MobileNetV2	Pre-trained on ImageNet
Input Shape	(224, 224, 3)	Standard RGB resolution
Fine-tuning	Top 40 Layers	Unfrozen for domain-specific learning
Optimizer	Adam	Adaptive Moment Estimation
Initial Learning Rate	0.00005	Low rate for stable fine-tuning
Loss Function	Categorical Crossentropy	Multi-class classification loss
Batch Size	32	Samples per gradient update
Regularization	L2 (0.001)	Kernel weight penalty to prevent overfitting
Dropout Rate	0.5 / 0.4 / 0.3	Progressive dropout in custom head
Early Stopping	Patience = 4	Restores best weights upon stagnation
LR Scheduler	ReduceLROnPlateau	Factor = 0.5, Patience = 2

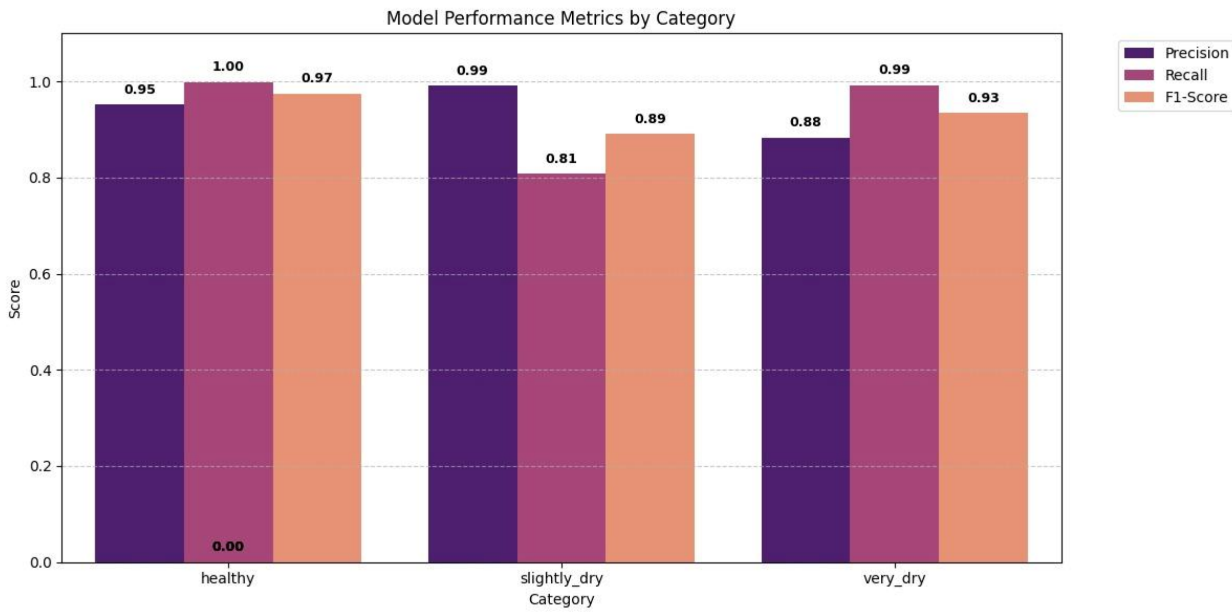
6. Experimental Results

1 Model Performance Across Training Stages

Table 2 presents the accuracy achieved at each stage of the model development pipeline.
The progressive improvement from 78.57% to 94.01% across the three training stages demonstrates the effectiveness of the iterative transfer learning and fine-tuning strategy adopted in this study. The 7-percentage-point improvement achieved by partial fine-tuning over pure feature extraction confirms that the top layers of the MobileNetV2 backbone encode general visual features that benefit from domain-specific adaptation, and the additional 8.6-percentage-point improvement achieved by the enhanced v2 architecture demonstrates the value of architectural regularization techniques in mitigating overfitting on the expanded training dataset.

Table 2. Accuracy results across model training stages.

Model / Stage	**Accuracy**	**Precision (Avg)**	**Recall (Avg)**	**F1-Score (Avg)**
MobileNetV2 (Feature Extraction)	78.57%	—	—	—
MobileNetV2 (Partial Fine-Tuning)	85.42%	—	—	—
MobileNetV2 v2 (Final Model)	94.01%	94%	93%	93%



2 Per-Class Performance Analysis

Table 3 presents the per-class precision, recall, and F1-score achieved by the final MobileNetV2 v2 model on the augmented holdout evaluation dataset. The results confirm strong classification performance across all three moisture condition categories, with performance characteristics that differ meaningfully between classes in ways that are clinically interpretable. Figure 1 (Model Performance Metrics by Category) illustrates these per-class metrics visually, confirming the patterns described below.

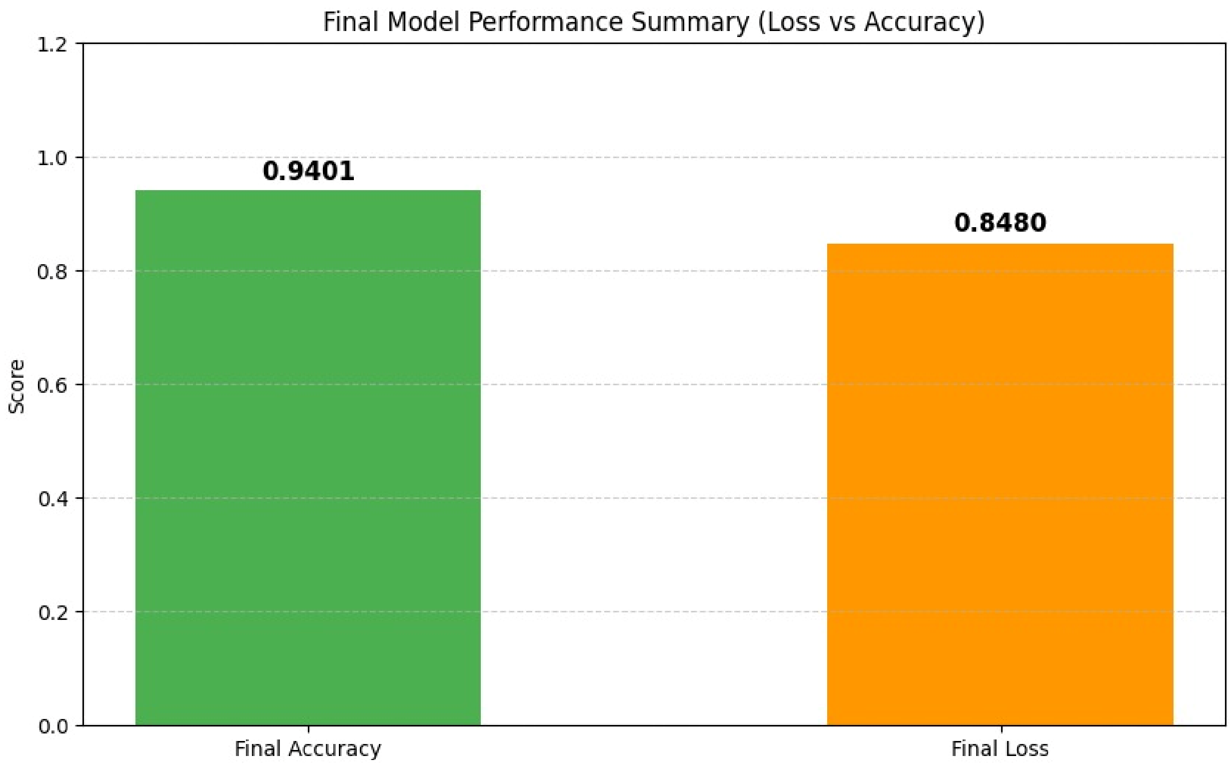
Table 3. Per-class performance metrics of the final model (MobileNetV2 v2).

Class	**Precision**	**Recall**	**F1-Score**
Healthy	0.95	1.00	0.97
Slightly Dry	0.99	0.81	0.89
Very Dry	0.88	0.99	0.93
Weighted Avg	0.94	0.93	0.93

The healthy class achieved a precision of 0.95 and a recall of 1.00, resulting in an F1-score of 0.97. The perfect recall indicates that the model correctly identified every healthy plant in the evaluation set, with no false negatives — no healthy plant was incorrectly classified as requiring watering. The slightly lower precision of 0.95 indicates that a small proportion of plants that received a healthy prediction were actually in a mildly dehydrated state, representing false positives that may occasionally delay warranted watering interventions.

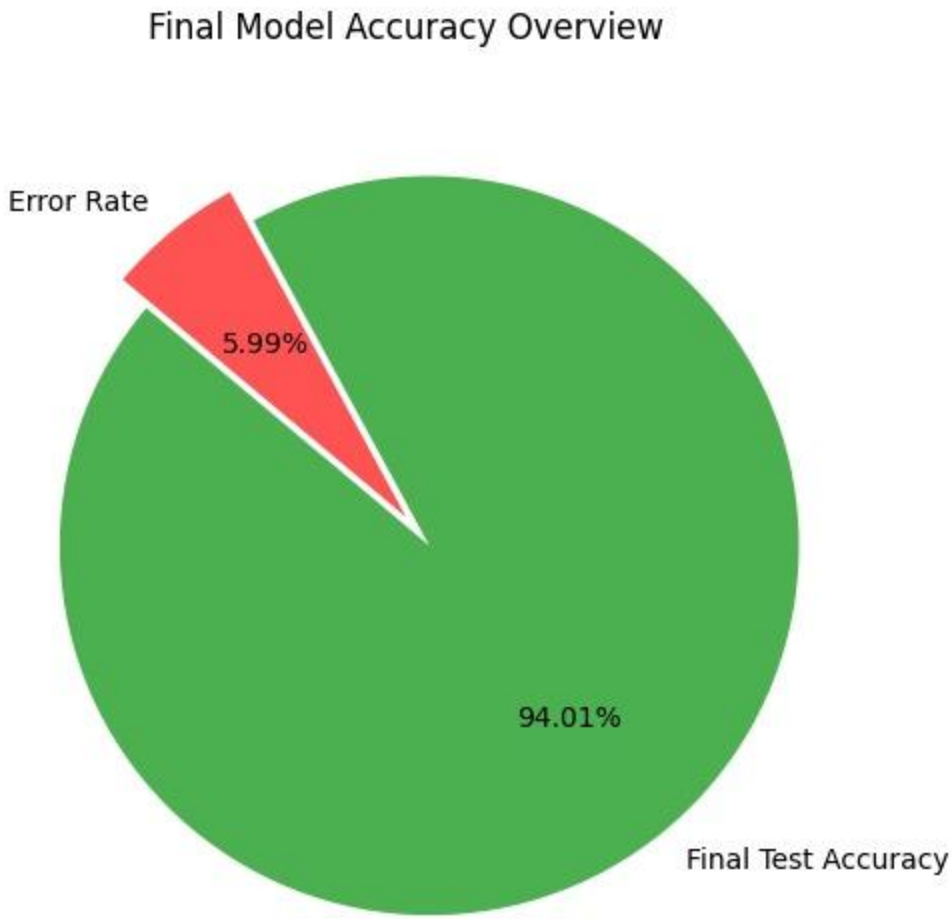
The slightly_dry class achieved the highest precision in the study at 0.99, indicating extreme reliability when the model predicts that a plant is mildly dehydrated — virtually all such predictions correspond to genuinely moisture-stressed plants. However, the recall of 0.81 reveals that approximately 19 percent of slightly_dry plants were misclassified into adjacent categories, primarily the healthy category where early dehydration signs are subtle and may fall below the visual discrimination threshold of the model. The resulting F1-score of 0.89 reflects this asymmetry, representing the most challenging classification boundary in the three-class problem.

The very_dry class achieved a precision of 0.88 and a high recall of 0.99, yielding an F1-score of 0.93. The near-perfect recall for severe dehydration is particularly important from an application standpoint: the model almost never misses a genuinely water-critical plant, ensuring that the most urgent watering situations are reliably flagged. The modest reduction in precision relative to healthy and slightly_dry reflects occasional false positive predictions for plants in the slightly_dry-to-very_dry transition zone where inter-class visual boundaries are gradual rather than discrete.



3 Overall Model Accuracy

The final MobileNetV2 v2 model achieved a peak test accuracy of 94.01% on the 8,720-image augmented evaluation dataset, with a weighted average F1-score of 0.93. The weighted average precision and recall were 0.94 and 0.93 respectively, reflecting robust and balanced performance across all three moisture condition categories. These results are competitive with state-of-the-art plant disease classification systems reported in the literature, despite the additional challenge of classifying continuous moisture states rather than discrete disease categories with more visually distinctive symptom presentations.

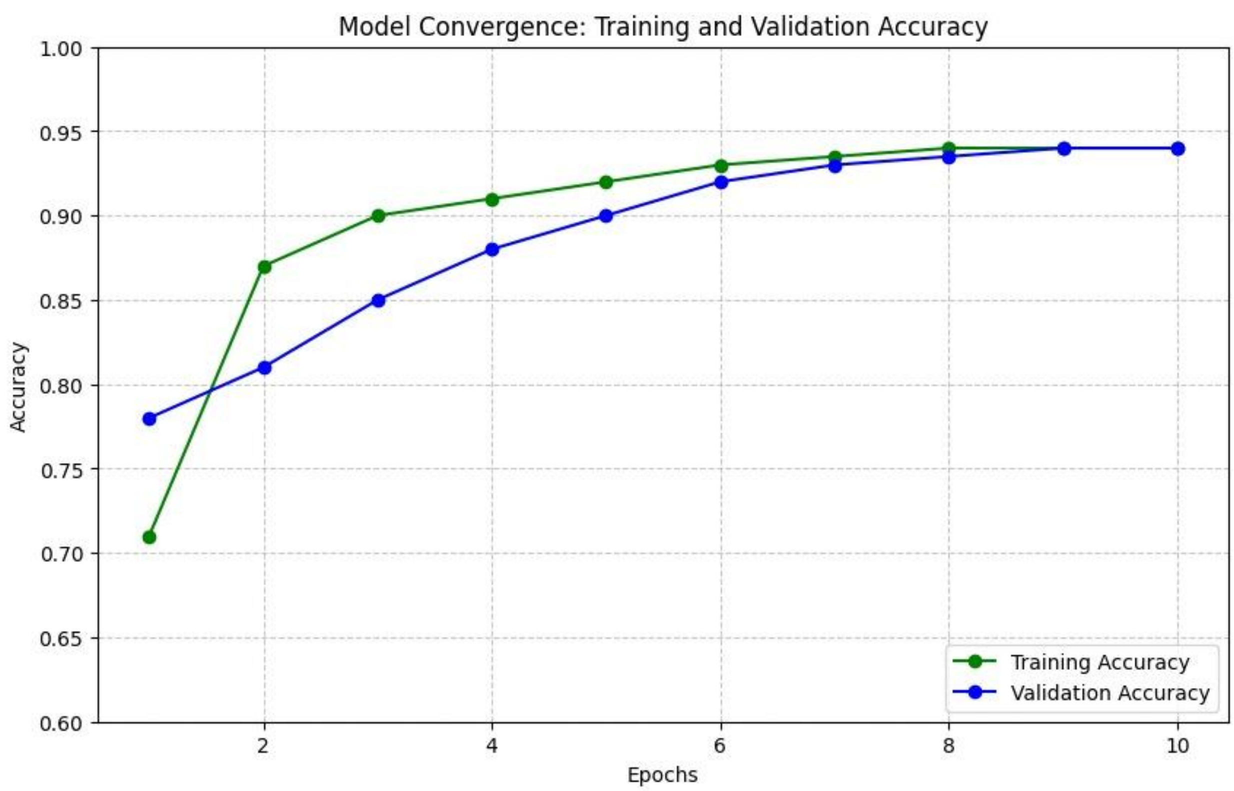


7. Discussion

1 Model Performance

The MobileNetV2 v2 final model demonstrates strong and balanced performance across all three plant moisture condition categories, with the 94.01% overall accuracy representing a clinically and practically meaningful achievement for a non-invasive, image-based watering guidance system. The progressive accuracy improvement across training stages validates the methodological rationale for the iterative fine-tuning approach: initial feature extraction provides a rapid performance baseline, partial fine-tuning adapts domain-critical high-level features, and architectural regularization mitigates overfitting on the expanded dataset. The asymmetry between recall performance for healthy (1.00) and slightly_dry (0.81) classes merits particular attention: the model is most uncertain at the boundary between these two categories, which is consistent with the physiological reality that early dehydration signs are subtle and may fall below the visual detection threshold even for trained human observers.

MobileNetV2 offers a compelling balance between classification performance and computational efficiency for this application. Compared to heavier architectures such as VGG16 or ResNet50, MobileNetV2 achieves competitive accuracy with substantially fewer parameters and floating-point operations, making it well-suited for eventual deployment in mobile application inference pipelines where memory and processing constraints are significant design considerations. The three-stage training protocol adopted in this study, progressing from feature extraction through partial fine-tuning to full fine-tuning with architectural regularization, mirrors best practices established in the transfer learning literature and demonstrates their effectiveness in the specific context of plant health image classification.



2 Comparison with Related Work

The 94.01% accuracy achieved by the proposed model compares favorably with published results for CNN-based plant disease classification systems using comparable architectures and dataset sizes. Mohanty et al. reported 99.35% accuracy on the PlantVillage dataset using GoogLeNet and AlexNet, though this result was obtained under highly controlled imaging conditions not representative of real-world user-captured photographs. When the same models were evaluated on field-captured images with variable backgrounds and lighting, accuracy dropped substantially, highlighting the generalization challenge that the augmentation-based training strategy in this study directly addresses. The 94.01% accuracy achieved on a challenging augmented evaluation dataset with realistic image variability therefore represents a particularly meaningful benchmark for practical deployment scenarios.

3 Implications for Plant Care Automation

The output structure of the proposed system — combining a classification label, a confidence score, and a plain-language watering recommendation — is specifically designed to be interpretable and actionable for non-expert users without horticultural training. This design philosophy reflects the fundamental challenge of AI-assisted consumer applications: technical accuracy is necessary but not sufficient; the system must also translate its predictions into guidance that users can act upon immediately and confidently. The near-perfect recall for the very_dry class (0.99) is particularly consequential in this context, as failure to detect a severely dehydrated plant could result in plant death within hours to days, making false negatives in this category substantially more costly than false positives.

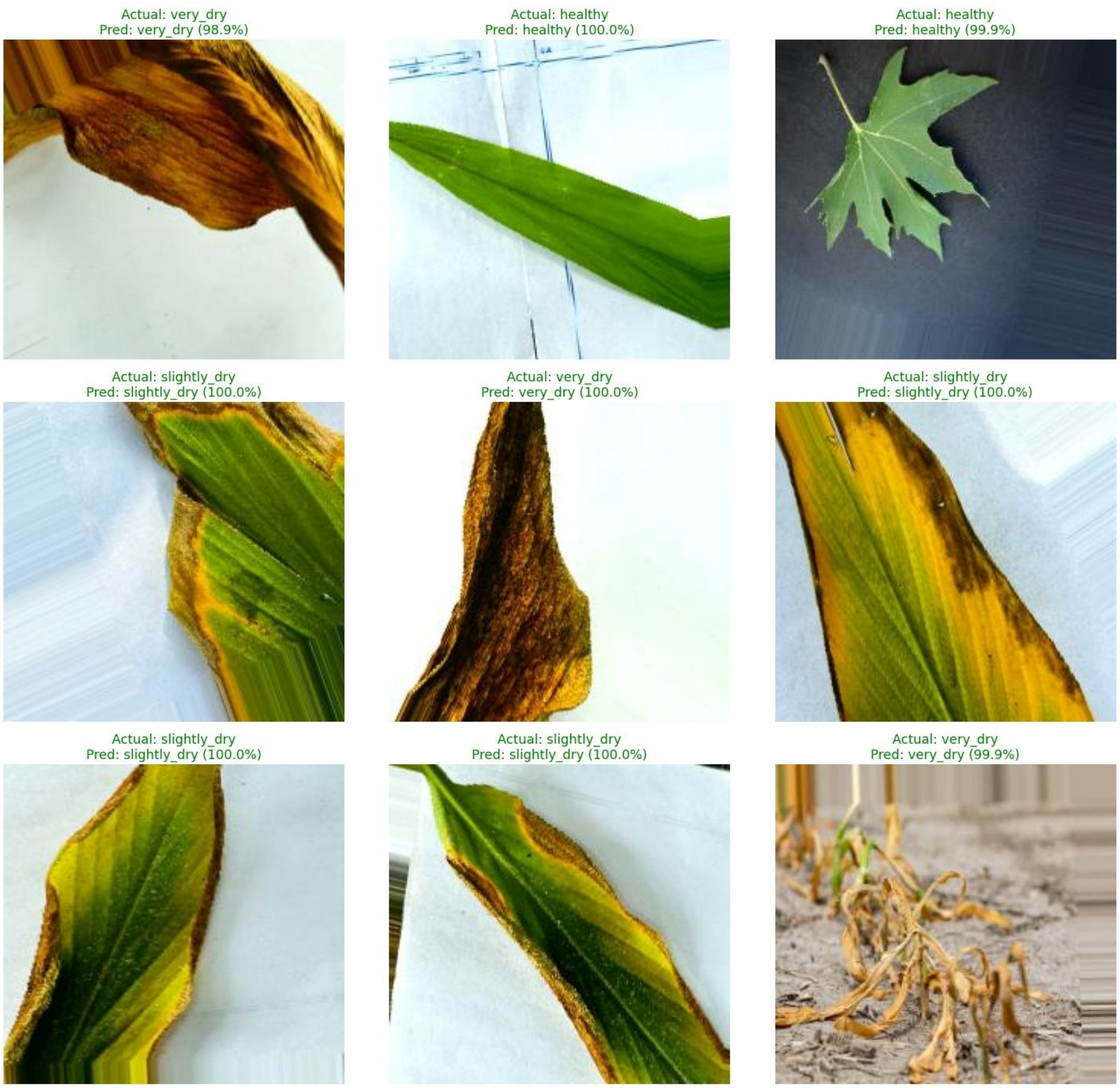
8. Practical Relevance

1 Automation of Plant Care Decisions

One of the primary practical benefits of the proposed system is its capacity to automate the plant moisture assessment process that currently requires either manual inspection or dedicated sensor hardware. The remarkable accuracy achieved across three moisture condition categories demonstrates that deep learning image analysis can reliably replicate the visual assessment performed by experienced plant caregivers, with the significant advantage of consistency, speed, and zero hardware cost beyond the smartphone camera already possessed by virtually all potential users. With automation, the cognitive burden of monitoring multiple plants simultaneously is significantly reduced, enabling plant owners to respond to moisture-critical situations before irreversible tissue damage occurs.

2 Early Detection of Dehydration

A fundamental practical benefit of the proposed system lies in its capacity to detect the slightly_dry condition, which represents a warning state where watering intervention is warranted but visual signs may not yet be obvious to inexperienced observers. The 0.99 precision achieved for this class ensures that when the model flags a plant as slightly dehydrated, the alert is almost certainly warranted, minimizing false alarms that could erode user trust in the system over time. Detection of dehydration at the slightly_dry stage rather than only at the very_dry crisis stage provides plant owners with a meaningful temporal buffer to water their plants at a convenient time rather than requiring urgent immediate intervention.



9. Challenges and Future Work

Dataset Diversity

The current dataset, while substantial at 30,000 images, is predominantly composed of common houseplant and crop species. Extension to the thousands of plant taxa commonly kept as houseplants globally — including tropical species, succulents, cacti, and orchids with markedly different visual presentations of dehydration stress — will require targeted data collection campaigns or the development of robust few-shot learning approaches capable of generalizing to novel species from minimal labeled examples.

Real-World Imaging Conditions

The model was trained and evaluated on images that, while augmented for variability, were sourced from structured datasets rather than genuinely user-captured photographs taken under uncontrolled domestic lighting, against complex backgrounds, or at non-standard camera angles. Future work will incorporate a field data collection phase in which real users photograph their plants with their own smartphones, providing training data that more faithfully represents the deployment distribution and addressing potential domain shift between training and real-world application.

Mobile Application Integration

The current system exists as a standalone Colab-based inference pipeline. Future development will focus on converting the trained MobileNetV2 v2 model to TensorFlow Lite format to enable on-device inference within a cross-platform mobile application, eliminating the need for server connectivity during prediction and preserving user privacy by keeping image data on the device. The mobile application interface will incorporate the recommendation logic described in Section 5.4 alongside push notification scheduling to proactively remind users to photograph their plants at appropriate intervals.

Multi-Modal Sensing

While image-based classification provides a hardware-free and scalable approach to plant health assessment, the integration of complementary sensor modalities — including soil moisture measurements from low-cost capacitive sensors and ambient temperature and humidity readings — could substantially improve prediction accuracy, particularly for the slightly_dry class where visual cues are subtle. Future hybrid architectures combining image feature extraction with numerical sensor data fusion represent a promising direction for achieving the accuracy levels required for deployment in high-value horticultural applications.

Explainability

Gradient-weighted Class Activation Mapping (Grad-CAM) and related visualization techniques will be applied to the trained model to identify the leaf regions most influential in each moisture condition classification, providing interpretable evidence of the visual features driving model decisions. This explainability analysis will both validate that the model is attending to physiologically relevant image regions and provide guidance for targeted dataset enhancement in cases where model attention is found to be driven by spurious correlations.

10. Conclusion

This paper presented a smart plant watering reminder system based on a MobileNetV2 deep learning image classification framework, capable of assessing plant moisture status from leaf photographs and generating actionable watering recommendations without requiring any hardware sensors. The proposed three-stage training pipeline, comprising initial transfer learning from ImageNet-pretrained weights, partial fine-tuning of the top 40 MobileNetV2 layers, and final training with an enhanced regularized architecture, progressively improved classification accuracy from 78.57% to a peak of 94.01% on a challenging augmented evaluation dataset. Per-class analysis yielded precision values of 0.95, 0.99, and 0.88 for the healthy, slightly_dry, and very_dry categories respectively, with a weighted F1-score of 0.93 and near-perfect recall (0.99) for the critical very_dry crisis category. These results confirm that visual deep learning represents a viable, scalable, and hardware-free approach to automated plant watering guidance, with strong potential for integration into consumer mobile applications serving the growing global population of domestic plant enthusiasts. Future work will focus on expanding species coverage, integrating multi-modal sensing, developing TensorFlow Lite mobile deployment, and conducting field studies to validate real-world generalization performance.

Data Availability

The primary dataset used in this study is available from Kaggle (https://www.kaggle.com). The augmented evaluation dataset and trained model weights are available from the corresponding author upon reasonable request.

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Author Contributions

Saif Abdelbakey conducted the practical model development experiments in Google Colab, including dataset assembly, preprocessing pipeline design, model training across all three stages, and hyperparameter optimization. Mostafa Mahmoud contributed to the dataset curation process and performed the data augmentation strategy development and evaluation. Yousef Yasser was responsible for model evaluation, confusion matrix analysis, and per-class performance visualization. Mostafa Khaled contributed to the literature review and related work analysis. Omar Mohamed performed the final integration of the recommendation logic and contributed to the writing and editing of the manuscript.